

## Search sheets: how to run the activity

LL Ms Un plug ged

The pages after this one are 15 **search sheets**, built from *Green Eggs and Ham* by Dr Seuss. Print them one-sided and hand out **one sheet per person** — every sheet is different, and nothing needs cutting out.

**If the room is a different size**, match it to the sheets rather than the other way round. More people than sheets: print the set twice and hand out both copies. Every token pair is then held by two people, so every count doubles and the proportions — which is all the model is — do not move. It buys some insurance too, because one person away no longer takes a token pair out of the room with them. Fewer people than sheets: give the spares out rather than leaving them on the table, two or three to a person if it comes to that. A sheet nobody holds is a hole in the model.

Each sheet holds 66–67 **token pairs**. A **token** is one word or one punctuation mark; a token pair is a **next token** paired with the **previous token** that came immediately before it somewhere in the original text.

In **and ham**, the boxed token is the previous token and *ham* is the next token. Each token keeps the same colour wherever it appears, so scan by colour first and read second. With only 8 colours and far more tokens than that, a colour narrows the search rather than settling it — always check the token itself.

Say the colour when you call a token out — “who has *ham*? it’s a red one” — and most of the page stops being worth reading. The swatches are chosen to answer to these words, so the words are the ones to use; the sheets themselves print no key, because a strip of swatches at the top of a page of swatches is one more thing to scan past.

**black red brown green teal blue purple magenta**

### Running a round

Seed the board first. Ask someone to read out any token pair from their sheet — both tokens, boxed one first — and write it up. It matters that the opening comes off a sheet rather than out of your head: the room supplies the beginning as well as everything after it. Then, each round:

1. **Call out the last token** of the text so far.
2. Everyone scans their own sheet for token pairs whose previous token matches. **Hands up** — one hand each, and anyone with more than one match just goes with the first they spot.
3. **Pick a raised hand at random** and ask for that token pair’s next token.
4. A scribe writes the token on the board. That token becomes part of the context for the next round. Repeat.

Picking at random matters. If you take whoever shouts first you are sampling the fastest reader, not the text — and the whole point is that the room samples the way the model does.

### What to watch for

**The hands are the probability distribution.** Every token pair was dealt to exactly one person, and the deal spreads each context across as many people as it will go, so a common continuation really does put more hands in the air than a rare one. Nobody has to explain weighted sampling; the room performs it. (Roughly: for a context more common than the class is big, somebody holds two matches but raises one hand, which flattens the busiest contexts.)

**Nobody holds the model.** No single sheet can continue the text on its own. The model only exists across the whole room — and if somebody is away, their share of it is missing, and some contexts will draw no hands at all.

**Quiet rounds.** One hand is not a stall — a word with a single continuation is the model working. Nor will you run dry: a word has nothing following it only if it appears nowhere but the very last position in the text, which is at most one word in the whole deal. So no hands at all almost always means an empty seat — two of them, if you printed the set twice — and that is worth saying out loud. Reseed from any token pair and carry on.

993 tokens, 56 unique · 1.8 bits/token · perplexity 3.5 · branching factor 3.45

## Worked example

**Start anywhere.** Pick any token pair — say this one — and write its previous token *and* its next token on the board.

**Round 1.** Call out the last token on the board (**ham**). Another sheet has a match, so that person raises a hand and reads out ..

**Round 2.** Call out the last token on the board (.). Another sheet has a match, so that person raises a hand and reads out *Sam*.

Keep going for as long as you like. The text the room writes will sound like *Green Eggs and Ham* without ever repeating it — that is the model generating, not the corpus being read out.

*the token pair that answers*

and ham

ham .

. Sam

*on the board*

and ham

and ham .

and ham . Sam

and ham

and ham

do not

a mouse

would not

I will

train .

are so

would eat

. .

! in

! I

. Sam

them anywhere

a car

I am

here or

them in

them here

. I

could not

Sam I

not in

, could

not ,

not like

in the

them anywhere

let me

ham .

do not

dark ?

I will

like green

with a

a train

not like

I do

, Sam

am .

! I

eat them

, could

you may

? would

. I

you like

tree !

on a

. I

or there

you like

like them

box .

there ?

not ,

a train

will eat

them with

do not

goat .

the dark

in a

in a

green eggs

. I

I do

|           |            |            |          |
|-----------|------------|------------|----------|
| in the    | like them  | , could    | and I    |
| them ,    | , with     | do not     | them in  |
| would not | a tree     | would not  | a train  |
| do not    | train !    | I will     | with a   |
| you eat   | there .    | in the     | and I    |
| not like  | tries them | not on     | I will   |
| you on    | them with  | eat them   | ? I      |
| . try     | green eggs | ! If       | a box    |
| ! a       | . I        | . not      | he tries |
| eat green | here or    | like green | eggs and |
| not ,     | ham .      | could not  | house .  |
| dark .    | not like   | not with   | you do   |
| I do      | . and      | . that     | on a     |
| or there  | tree .     | I will     | Sam !    |
| will not  | . I        | them here  | am .     |
| ! I       | with a     | , will     | do not   |
| like them | in the     | I do       |          |

|            |           |          |           |
|------------|-----------|----------|-----------|
| would not  | train .   | tree .   | do not    |
| green eggs | in a      | them ,   | do not    |
| do not     | a tree    | I would  | like them |
| . a        | ! so      | or there | I do      |
| , could    | not eat   | in a     | eggs and  |
| ? I        | would not | you !    | I do      |
| with a     | them here | house .  | could not |
| them ,     | dark !    | I will   | like them |
| am Sam     | Sam I     | and I    | them ,    |
| . .        | not like  | not like | you do    |
| . would    | will not  | , with   | there .   |
| with a     | eat them  | If you   | and there |
| not ,      | in a      | . I      | ! would   |
| . Sam      | like them | on a     | here and  |
| ! you      | , Sam     | mouse .  | . I       |
| not like   | eat them  | a house  | you ,     |
| a train    | I do      |          |           |

|           |               |           |            |
|-----------|---------------|-----------|------------|
| . and     | tree .        | train !   | house ?    |
| not eat   | I would       | would you | ! Thank    |
| , you     | car !         | . not     | a house    |
| am .      | them anywhere | will eat  | do not     |
| , on      | mouse .       | . I       | I would    |
| . they    | Sam I         | them in   | eat them   |
| I do      | them with     | eggs and  | green eggs |
| you eat   | ! I           | not ,     | eat them   |
| not in    | with a        | a house   | do not     |
| you see   | . say         | say .     | there .    |
| a car     | here or       | like them | them in    |
| in a      | like them     | with a    | , Sam      |
| would not | like them     | Sam I     | you ,      |
| dark .    | not like      | Thank you | a box      |
| not eat   | in a          | on a      | I do       |
| . you     | I would       | and ham   | in a       |
| ? eat     | could you     |           |            |

|            |            |           |           |
|------------|------------|-----------|-----------|
| them in    | a box      | not like  | not like  |
| not in     | them ,     | there .   | I will    |
| green eggs | I do       | eggs and  | a mouse   |
| I would    | not like   | will eat  | I would   |
| in a       | I do       | . say     | Sam !     |
| anywhere . | Thank you  | Sam ,     | like them |
| here in    | not like   | in a      | . I       |
| with a     | am !       | would not | would you |
| like them  | . would    | them with | say !     |
| do not     | tree !     | train !   | eat them  |
| a mouse    | car !      | house .   | mouse .   |
| . I        | like green | , you     | you may   |
| . .        | a boat     | not in    | with a    |
| on a       | ! you      | , Sam     | do not    |
| and ham    | could not  | . could   | ? not     |
| ! here     | boat .     | them !    | eat them  |
| you ,      | dark ?     |           |           |

|            |               |               |            |
|------------|---------------|---------------|------------|
| in a       | Sam I         | a tree        | I am       |
| the dark   | them anywhere | like them     | car .      |
| . I        | will eat      | train !       | not like   |
| boat .     | you like      | so like       | . and      |
| a car      | like them     | , could       | Sam I      |
| do not     | not like      | them anywhere | do not     |
| eat them   | not like      | . I           | am .       |
| . I        | tree !        | you see       | , in       |
| eggs and   | say .         | . would       | ! not      |
| house .    | would not     | not eat       | them .     |
| I would    | may .         | ? not         | . I        |
| eat them   | not ,         | dark .        | would you  |
| in a       | mouse ?       | there .       | them with  |
| green eggs | a car         | I do          | with a     |
| ! a        | not on        | and ham       | I will     |
| on a       | a tree        | I would       | like green |
| could you  | anywhere .    |               |            |



like them

like green

do not

I do

I will

! a

! I

. not

Sam I

do not

dark !

I do

boat ?

them !

not like

mouse .

a car

you ,

you may

in a

them here

in a

not like

not like

would you

could you

, could

green eggs

am !

eat them

. I

not in

car ?

ham !

house .

not like

eggs and

train ?

a fox

I am

and I

? I

anywhere .

them in

there .

. I

. I

I do

like green

may ,

. I

with a

so good

. .

the dark

a house

will not

say !

not in

them ,

eat them

would you

Sam I

I say

, on

a mouse

|           |            |            |            |
|-----------|------------|------------|------------|
| them with | ? could    | eggs and   | not like   |
| not like  | anywhere . | ! I        | car .      |
| not like  | ham ?      | not ,      | would you  |
| you ,     | , in       | in a       | . I        |
| mouse .   | house .    | say !      | green eggs |
| will eat  | . I        | fox .      | like them  |
| I do      | a box      | them here  | , with     |
| I do      | could not  | a boat     | with a     |
| there .   | a house    | . I        | and in     |
| them .    | am .       | . I        | may like   |
| . I       | you ,      | I will     | would you  |
| ! not     | I like     | like green | and in     |
| . I       | eat green  | do not     | them !     |
| train !   | so you     | eat them   | the rain   |
| I will    | . would    | like green | I do       |
| in the    | do !       | not ,      | a train    |
| not eat   | Sam .      |            |            |

|          |           |            |            |
|----------|-----------|------------|------------|
| I do     | . not     | in the     | ? I        |
| , Sam    | fox .     | eat them   | I am       |
| there .  | house .   | would not  | a box      |
| not eat  | you ,     | with a     | a boat     |
| not like | am .      | them in    | . and      |
| and you  | eggs and  | could not  | rain ?     |
| car !    | do not    | the rain   | like them  |
| not like | . you     | I would    | . not      |
| not like | like that | them in    | anywhere . |
| in a     | Sam .     | like green | would you  |
| ! try    | will eat  | eat them   | try them   |
| I am     | not like  | so good    | I do       |
| I am     | may like  | you ?      | not in     |
| . try    | ! I       | mouse .    | them ,     |
| a goat   | and ham   | do not     | them in    |
| . I      | , Sam     | ham !      | green eggs |
| a fox    | . I       |            |            |

|            |           |           |            |
|------------|-----------|-----------|------------|
| .you       | a tree    | a fox     | them .     |
| eat them   | a goat    | the dark  | will try   |
| not in     | them in   | fox .     | car .      |
| here they  | try them  | eggs and  | not in     |
| , so       | ham .     | them with | you will   |
| , in       | I would   | Sam I     | . not      |
| in a       | not like  | I do      | in the     |
| green eggs | ! would   | in a      | eat them   |
| not with   | mouse .   | like them | do not     |
| . and      | , I       | ? I       | anywhere . |
| I do       | . I       | a fox     | rain .     |
| ! Sam      | with a    | I am      | be .       |
| not ,      | I am      | would eat | you may    |
| do not     | . I       | them here | not like   |
| so I       | box ?     | would you | like them  |
| and on     | like them | I am      | am Sam     |
| and in     | . I       |           |            |

|            |               |           |           |
|------------|---------------|-----------|-----------|
| , in       | , in          | box .     | not in    |
| a house    | I do          | not in    | am .      |
| and ham    | not like      | ! eat     | with a    |
| fox .      | will eat      | in a      | try them  |
| be !       | eat them      | am !      | would you |
| rain .     | anywhere !    | them in   | not ,     |
| green eggs | and you       | like them | I do      |
| I am       | . I           | I do      | ham .     |
| or there   | not like      | them in   | you let   |
| them with  | you ?         | like them | ? in      |
| could you  | . I           | a goat    | a train   |
| do not     | the dark      | do not    | I like    |
| not like   | . and         | in a      | here or   |
| . not      | eggs and      | I will    | . I       |
| ? I        | a goat        | in a      | . and     |
| would you  | like them     | , could   | ! Thank   |
| Sam I      | them anywhere |           |           |

will see

ham .

a fox

them in

be !

box .

. you

. and

or there

! you

could you

. I

not eat

! could

I am

fox .

not like

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them in

not eat

them !

try them

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. not

anywhere !

in the

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. and

you let

I do

and ham

would not

eat them

a fox

? would

I will

, Sam

, Sam

. you

, would

am !

Sam I

do not

a house

the dark

and I

rain .

I do

eggs and

will not

do not

not ,

like them

a car

them ,

! and

with a

you ,

not with

I am

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here or

I do

see .

like them

will not

not ,

do not

in a

in the

I will

I will

the dark

them with

a house

them and

see .

eat them

. not

ham ?

! let

not like

me be

. I

not ,

they are

a train

! that

a fox

goat !

, would

be ,

that Sam

let me

like them

! and

and ham

you ,

with a

Sam I

would not

them .

in a

I would

here or

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like them

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not ,

a mouse

do not

you ,

I am

. not

fox .

, could

and ham

, could

I am

box .

|               |           |           |           |
|---------------|-----------|-----------|-----------|
| not eat       | them with | or there  | , Sam     |
| could you     | a tree    | I would   | not like  |
| box .         | you like  | , would   | not like  |
| goat ?        | ! I       | I do      | here or   |
| ham .         | . I       | I am      | ? here    |
| like them     | the rain  | like them | like them |
| not with      | . I       | fox ?     | ! would   |
| . would       | . and     | a mouse   | they are  |
| that Sam      | let me    | you let   | them ,    |
| in a          | see .     | me be     | I will    |
| . say         | Sam I     | and ham   | not eat   |
| them anywhere | them ,    | a mouse   | I do      |
| will not      | . not     | and I     | do not    |
| would not     | a box     | will not  | do not    |
| do not        | in a      | in the    | good ,    |
| I do          | ! I       | eat them  | with a    |
| am .          | , Sam     |           |           |



let me

not in

them here

I am

a box

I would

. and

will not

and he

I do

in a

you say

you will

like them

would not

. I

goat .

I could

I do

do not

will see

! I

them anywhere

a train

not in

with a

, Sam

in the

, could

good ,

them here

are .

not with

. not

do not

! not

see !

I do

would you

box .

am .

. .

and I

a car

here or

do not

? I

ham !

like them

not eat

. not

eat them

! not

them here

, I

or there

a mouse

Sam I

that Sam

not in

the rain

in a

like them

could not

. I

me be